

Employee Attrition Prediction Using Machine Learning

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Abstract— Employee attrition prediction is an important task in human resource analytics aimed at reducing employee turnover and improving organizational stability. This project focuses on developing a machine learning-based system to predict whether an employee is likely to leave the organization. The approach involves data preprocessing, feature engineering, and model building using algorithms such as Logistic Regression, Random Forest, and XGBoost. A dataset containing 10,000 employee records is used, and additional features like income per year, work stress, and promotion gap are created to enhance prediction capability. The models are trained and evaluated, where XGBoost achieves the best performance with an accuracy of 79.6%. Furthermore, a real-time prediction system is developed using Streamlit, allowing users to input employee details and obtain instant predictions, making the solution practical for real-world HR applications.

Index Terms— Machine Learning, XGBoost, Random Forest, Logistic Regression, Feature Engineering, SMOTE, HR Analytics, Streamlit, Real-Time Prediction System.

I. INTRODUCTION

Employee attrition refers to the reduction of employees in an organization due to resignation, retirement, or other factors. High attrition rates can negatively impact an organization by increasing recruitment costs, reducing productivity, and causing loss of experienced workforce. Therefore, understanding and predicting employee attrition has become a critical aspect of human resource management.

In recent years, the use of machine learning techniques in HR analytics has gained significant attention due to their ability to analyse large datasets and identify complex patterns. Traditional methods often fail to capture the underlying relationships between factors such as job satisfaction, work-life balance, income, and employee behaviour. Machine learning models, however, can effectively handle such complexities and provide accurate predictions.

This project aims to develop a predictive model for employee attrition using a structured dataset containing employee-related features. The study involves data preprocessing, feature engineering, and the application of multiple machine learning algorithms to identify the most effective model. Special emphasis is placed on deriving new features that capture hidden behavioural patterns influencing attrition.

In addition to prediction, the project also focuses on practical implementation by developing a real-time system that allows users to input employee details and receive immediate predictions. This enhances the usability of the model and makes it suitable for real-world organizational use. The overall objective of this work is to assist organizations in identifying potential attrition risks and taking proactive measures to improve employee retention.

This work proposes as follows:

- i. Data preprocessing and feature engineering are applied to clean the employee dataset and derive meaningful features such as work stress, promotion gap, and income per year.
- ii. Machine learning models such as Logistic Regression, Random Forest, and XGBoost are implemented to predict employee attrition effectively.
- iii. A Streamlit-based real-time prediction system is developed to allow users to input employee details and obtain instant attrition predictions.

II. LITERATURE REVIEW

Recent studies in employee attrition prediction have extensively utilized machine learning techniques to analyze workforce behavior and identify factors leading to employee turnover. In [1], the authors applied Logistic Regression and Decision Tree models to predict employee attrition using HR datasets. Their study demonstrated that basic classification models can achieve reasonable performance; however, they struggled to capture complex relationships among features such as job satisfaction and work environment. The research highlighted the need for more advanced models to improve prediction accuracy.

In another study, Random Forest and ensemble-based approaches were explored for attrition prediction [2]. The authors emphasized that ensemble methods improve prediction performance by combining multiple decision trees, thereby reducing overfitting and increasing robustness. While the results showed improved accuracy compared to traditional methods, the model complexity and computational cost were significantly higher, making real-time implementation challenging.

Deep learning techniques have also been introduced in HR analytics to enhance prediction capabilities. In [3], Artificial Neural Networks (ANN) were used to model employee behavior and predict attrition. The study achieved high accuracy but required large datasets and extensive training time. Additionally, the lack of interpretability in deep learning models made it difficult for HR professionals to understand the decision-making process.

Feature engineering has been identified as a crucial factor in improving model performance. In [4], the authors introduced derived features such as work stress and promotion gap to better represent employee behavior. Their results showed that engineered features significantly improved model accuracy. However, the study lacked real-time deployment, limiting its practical applicability in organizational environments.

More recent works have focused on combining machine learning models with real-time systems. In [5], XGBoost was applied for employee attrition prediction, achieving high accuracy due to its ability to handle complex data patterns. The study also emphasized feature

importance analysis for better interpretability. Despite its strong performance, the research did not fully address deployment challenges or usability in real-world HR systems.

Recent advancements in gradient boosting techniques have significantly improved the performance of employee attrition prediction systems. In [6], CatBoost and LightGBM models were applied to HR datasets to handle categorical features more efficiently and reduce preprocessing complexity. The study reported improved prediction accuracy and faster training compared to traditional ensemble methods. However, the research primarily focused on performance metrics and lacked deployment strategies for practical organizational use. The study suggested integrating lightweight deployment frameworks for real-time applications as future work.

In [7], support vector machines and hybrid machine learning approaches were utilized to analyse employee behaviour and predict attrition. The research highlighted that combining multiple algorithms improves prediction stability and reduces classification errors. Although the hybrid approach achieved

good performance, the model required extensive parameter tuning and higher computational resources. The authors identified the need for simplified yet effective systems that can balance prediction performance with usability in real-world HR environments.

Another study presented in [8] focused on explainable AI techniques for employee attrition prediction using feature importance analysis and interpretable machine learning models. The research emphasized the importance of transparency in HR decision-making and demonstrated how factors such as work-life balance, overtime, and job satisfaction influence employee turnover. While the study improved interpretability, the proposed system lacked real-time prediction capability and user interaction modules. Future research was directed toward integrating explainable models with deployable web-based systems for organizational applications.

From the above studies, it is evident that while many models achieve high accuracy, there is a gap in balancing performance with real-time usability and deployment. This project addresses this gap by combining feature engineering, model comparison, and a Streamlit-based real-time prediction system.

Publish Year	Reference	Features	Classification	Merits	Demerits	Research Gap
2021	[1]	Basic HR features (Age, Income, Satisfaction)	Logistic Regression, Decision Tree	Simple implementation, easy to interpret	Low accuracy, cannot capture complex patterns	Need for advanced models
2022	[2]	HR features + derived metrics	Random Forest, Ensemble Models	Improved accuracy, reduced overfitting	High computational cost, complex model	Efficient and lightweight models
2023	[3]	Large-scale HR dataset	Artificial Neural Networks (ANN)	High prediction accuracy	Requires large data, low interpretability	Need for explainable models
2022	[4]	Engineered features (Work Stress, Promotion Gap)	Machine Learning Models	Better representation, improved accuracy	No real-time implementation	Integration with deployment systems
2024	[5]	Full HR dataset + feature importance	XGBoost	High accuracy, handles complex data	Complex tuning, no deployment focus	Real-time prediction system

Table 1. Methods used in recent works

III. BACKGROUND STUDY

In recent years, machine learning techniques for employee attrition prediction have gained significant attention in the field of Human Resource Analytics. Organizations are increasingly utilizing predictive analytics to identify employees who are likely to leave the company and to improve employee retention strategies. Traditional methods mainly relied on statistical analysis and manual HR evaluation, which often failed to capture complex relationships among employee-related factors. Machine learning models provide a more efficient and data-driven approach by analyzing employee behavior patterns, job satisfaction, work-life balance, income, and organizational factors to predict attrition accurately.

Employee attrition prediction systems generally utilize structured HR datasets containing demographic, professional, and behavioral attributes of employees. Features such as monthly income, years at company, overtime, job satisfaction, performance rating, and relationship with managers play an important role in determining employee turnover. However, traditional machine learning systems often face challenges such as class imbalance, noisy data, and insufficient feature representation, which can reduce prediction performance and model reliability.

A. Drawback of Traditional HR Analysis

Traditional HR analysis methods mainly depend on manual observations, employee surveys, and statistical reporting techniques. These approaches are time-consuming and may not effectively capture hidden patterns within large-scale organizational data. In many cases, HR professionals are unable to identify employees at risk of attrition until resignation occurs, leading to increased recruitment costs and productivity loss.

Another major limitation of traditional methods is the inability to process high-dimensional employee data efficiently. Since employee behaviour depends on multiple interconnected factors, conventional analysis techniques may fail to capture complex relationships among features. Moreover, class imbalance in attrition datasets often leads to biased predictions where models tend to favour the majority class.

B. Importance of Feature Engineering

Feature engineering plays a crucial role in improving the performance of employee attrition prediction systems. Derived features such as income per year, promotion gap, and work stress help in capturing deeper employee behaviour patterns that are not directly available in the original dataset. These engineered features improve the learning capability of machine learning models and enhance prediction accuracy.

For example, employees with high workload and low job satisfaction may show a higher probability of attrition. Similarly, employees with long promotion gaps may experience reduced motivation and engagement. By creating meaningful features from existing attributes, the prediction system becomes more effective in identifying attrition risks.

C. Comparison between Logistic Regression, Random Forest, and XGBoost

Logistic Regression is one of the most widely used machine learning algorithms for binary classification problems such as employee attrition prediction. It is simple, interpretable, and computationally efficient. However, Logistic Regression may struggle to capture non-linear relationships among employee attributes, leading to lower prediction performance in complex datasets.

Random Forest improves prediction capability by combining multiple decision trees to reduce overfitting and increase robustness. It performs well on structured HR datasets and can identify important features influencing employee attrition. Nevertheless, Random Forest may require higher computational resources and longer training time compared to simpler models.

XGBoost is an advanced boosting algorithm designed to improve predictive performance through sequential learning. It effectively handles complex relationships, feature interactions, and class imbalance problems. Compared to Logistic Regression and Random Forest, XGBoost generally achieves higher accuracy and better generalization performance, making it suitable for employee attrition prediction tasks.

D. Importance of SMOTE for Class Balancing

Employee attrition datasets are usually imbalanced because the number of employees staying in the organization is significantly higher than the number of employees leaving. This imbalance can negatively affect machine learning performance by causing models to favour the majority class.

Synthetic Minority Oversampling Technique (SMOTE) is used to address this issue by generating synthetic samples for the minority class. SMOTE improves the balance between attrition and non-attrition classes, enabling machine learning models to learn patterns more effectively. As a result, the prediction system becomes more reliable and capable of identifying employees at risk of leaving the organization.

E. Accuracy

Accuracy in employee attrition prediction refers to the proportion of correctly classified employee records out of the total number of records in the dataset. It is calculated using the following formula:
$$\text{Accuracy} = (\text{Number of correctly Predicted Records}) / (\text{Total number of Records}) \times 100$$

Where, the number of correctly predicted records represents the count of employee instances accurately classified as attrition or non-attrition by the machine learning model. The total number of records includes all employee samples used during testing. Higher accuracy indicates better model performance and prediction capability.

IV. PROPOSED METHODOLOGY

4.1 Overview:

The proposed employee attrition prediction system is designed to analyse employee-related attributes and predict whether an employee is likely to leave the organization. The overall framework consists of data preprocessing, feature engineering, class balancing, machine learning-based classification, and real-time prediction deployment.

Initially, the employee dataset is collected and pre-processed to remove inconsistencies and convert categorical variables into numerical form. Feature engineering techniques are then applied to derive meaningful behavioural features such as income per year, work stress, and promotion gap. Since the dataset contains imbalanced classes, Synthetic Minority Oversampling Technique (SMOTE) is used to balance attrition and non-attrition samples.

After preprocessing, multiple machine learning models including Logistic Regression (LR), Random Forest (RF), and Extreme Gradient Boosting (XGBoost) are trained and evaluated. Among all models, XGBoost achieved the highest performance and was selected as the final classifier. Finally, the trained model is deployed using a Streamlit-based web application to provide real-time employee attrition prediction.

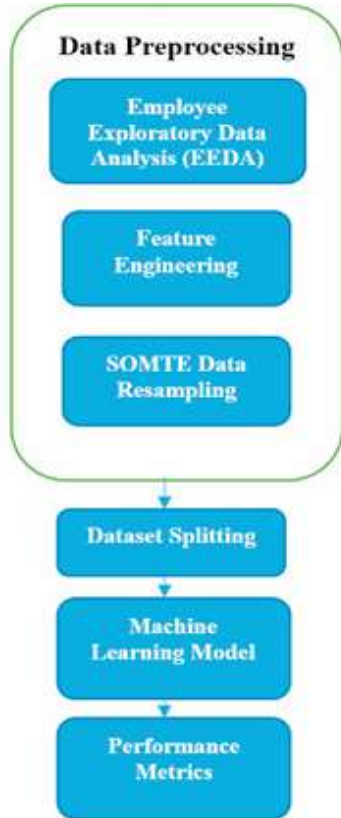


Fig. 1. Methodological analysis of the employee attrition prediction.

4.2 Data Preprocessing:

Data preprocessing is performed to improve dataset quality and ensure effective model training. Initially, the dataset is checked for missing values, duplicate records, and inconsistent entries. Since the dataset does not contain null values, no missing value imputation is required.

Categorical features such as gender, marital status, department, job role, and overtime are converted into numerical representations using label encoding techniques. Numerical features are standardized using StandardScaler to normalize the feature distribution and improve learning performance.

4.3 Feature Engineering:

Feature engineering is applied to derive additional meaningful features that better represent employee behavior and organizational patterns.

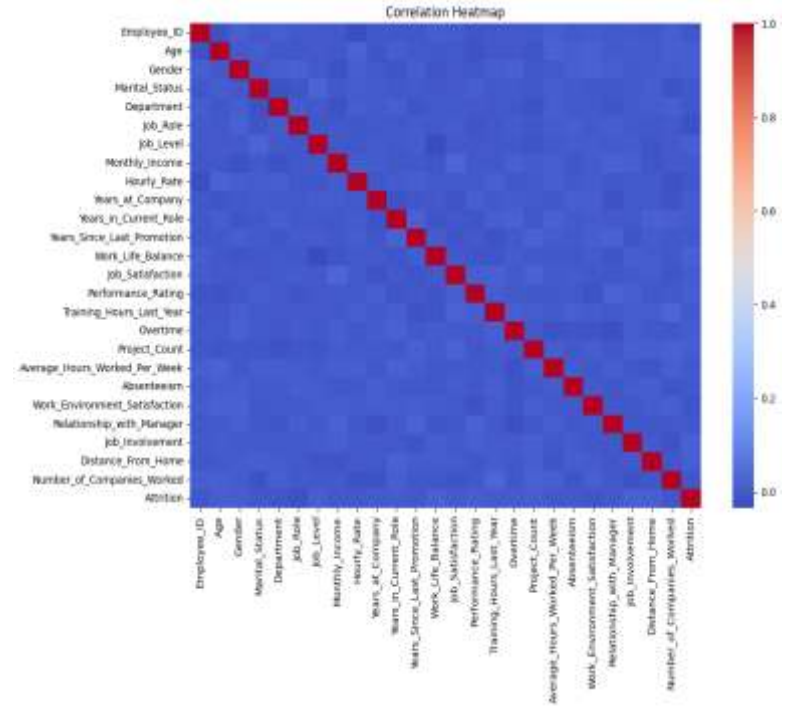


Fig. 2. The correlation analysis of dataset for feature engineering.

i. Income per Year

The income per year feature measures employee compensation relative to experience in the organization.

$$\text{Income_per_Year} = \frac{\text{Monthly_Income}}{\text{Years_at_Company} + 1}$$

ii. Work Stress

Work stress is estimated using employee workload and working hours.

$$\text{Work_Stress} = \text{Project_Count} \times \text{Average_Hours_Worked_Per_Week}$$

iii. Promotion Gap

Promotion gap represents the delay between employee experience and promotion history.

$$\text{Promotion_Gap} = \text{Years_at_Company} - \text{Years_Since_Last_Promotion}$$

These engineered features help improve prediction performance by capturing hidden employee behavior patterns.

4.3 SMOTE-Based Class Balancing:

Employee attrition datasets are generally imbalanced because the number of employees staying in the organization is significantly higher

than employees leaving. This imbalance negatively affects classification performance.

To overcome this issue, Synthetic Minority Oversampling Technique (SMOTE) is applied to generate synthetic samples for the minority class. SMOTE improves the learning capability of machine learning models and reduces prediction bias. The synthetic sample generation process is expressed as:

$$X_{new} = X_i + \lambda(X_{nearest} - X_i)$$

Where:

- X_i = minority class sample
- $X_{nearest}$ = nearest neighboring sample

λ = random value between 0 and 1

4.4 Machine Learning Models:

i. Logistic Regression (LR)

Logistic Regression is used as a baseline binary classification model to predict employee attrition probability. The sigmoid function used in Logistic Regression is given by:

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

Where: $z = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$

ii. Random Forest (RF)

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. The final prediction is obtained through majority voting:

$$RF(x) = \text{Majority Vote}(T_1(x), T_2(x), \dots, T_n(x))$$

Where:

- $T_n(x)$ represents the prediction of each decision tree.

iii. Extreme Gradient Boosting (XGBoost)

XGBoost is an advanced boosting algorithm that sequentially improves weak learners to minimize prediction errors. The objective function of XGBoost is defined as:

$$Obj = \sum l(y_i, \hat{y}_i) + \sum \Omega(f_k)$$

Where:

- $l(y_i, \hat{y}_i)$ = loss function
- $\Omega(f_k)$ = regularization term

XGBoost achieved the highest prediction accuracy among all implemented models.

V. RESULTS AND DISCUSSION

A. Overview of Experimental Results

The proposed employee attrition prediction system was evaluated using multiple machine learning models to determine the most effective classifier for predicting employee turnover. The experiments were conducted using a dataset containing 10,000 employee records with various

demographic, behavioural, and organizational attributes.

Initially, the dataset was pre-processed and standardized to improve model learning performance. Feature engineering techniques were applied to derive additional features such as income per year, work stress, and promotion gap. Furthermore, SMOTE was used to balance the dataset and reduce classification bias caused by imbalanced classes.

The performance of Logistic Regression (LR), Random Forest (RF), and XGBoost models was evaluated using accuracy metrics. Among the implemented models, XGBoost achieved the highest prediction accuracy, demonstrating its capability to capture complex relationships among employee attributes.

B. Dataset

The dataset used in this work contains employee-related information such as age, monthly income, job satisfaction, work-life balance, overtime, performance rating, and years at the company. The dataset initially consisted of imbalanced classes, where the number of non-attrition records was significantly higher than attrition records. After applying SMOTE, both classes were balanced effectively, improving the model's ability to learn attrition-related patterns. The dataset was divided into training and testing sets using an 80:20 ratio for model training and evaluation.

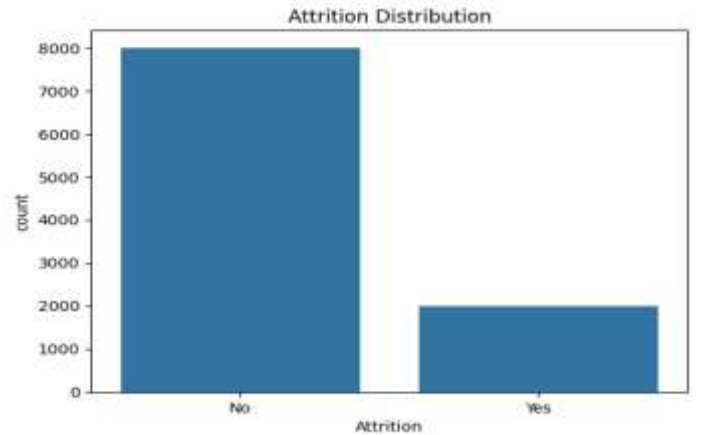


Fig. 3. Count of attrition values.

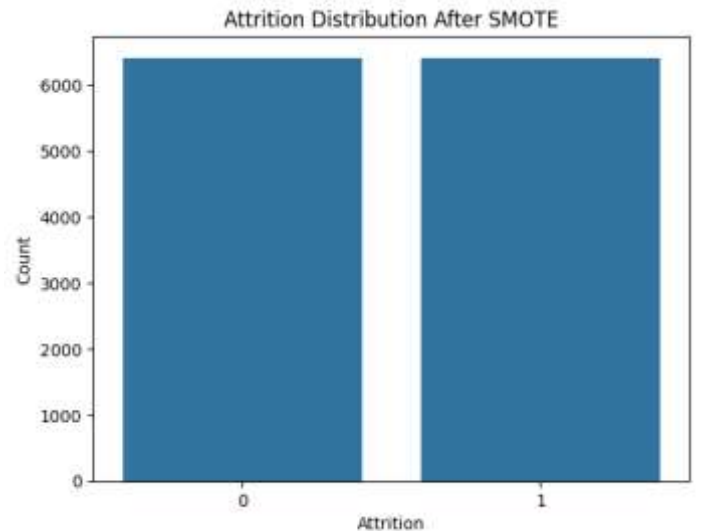


Fig. 4. Attrition Values After Applying SMOTE

C. Model Performance Analysis

Three machine learning algorithms were implemented and compared in this work.

Logistic Regression was used as the baseline classification model. Although the model is computationally efficient and interpretable, it showed lower performance in handling complex feature interactions within the employee dataset.

Random Forest improved prediction accuracy by combining multiple decision trees through ensemble learning. The model successfully reduced overfitting and provided better generalization compared to Logistic Regression.

XGBoost achieved the highest performance among all implemented models due to its boosting capability and efficient handling of feature interactions. The model effectively learned complex employee behavior patterns and produced better prediction results.

Model	Accuracy
Logistic Regression	49.05%
Random Forest	77.65%
XGBoost	79.60%

Table 2. Performance Comparison of Machine Learning Models

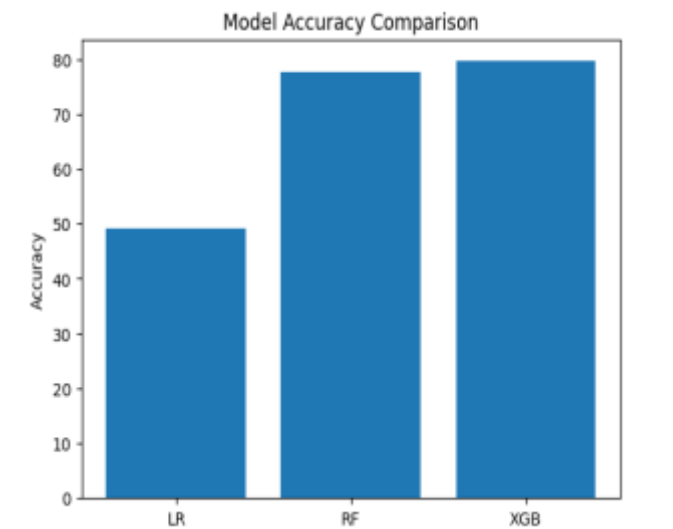


Fig. 5. Accuracy Comparison of Machine Learning Models

D. Confusion Matrix Analysis

The confusion matrix is used to evaluate the classification performance of the proposed XGBoost model by comparing actual employee attrition labels with predicted labels. It provides detailed information regarding correctly and incorrectly classified employee records.

The diagonal values in the confusion matrix represent correctly classified instances, while the off-diagonal values indicate misclassifications. The obtained confusion matrix demonstrates that the XGBoost model effectively

distinguishes between attrition and non-attrition classes with minimal classification errors.

The results indicate that the proposed model achieves strong predictive capability and balanced classification performance for both employee retention and attrition cases. This confirms the effectiveness of the proposed methodology in identifying employees at risk of leaving the organization.

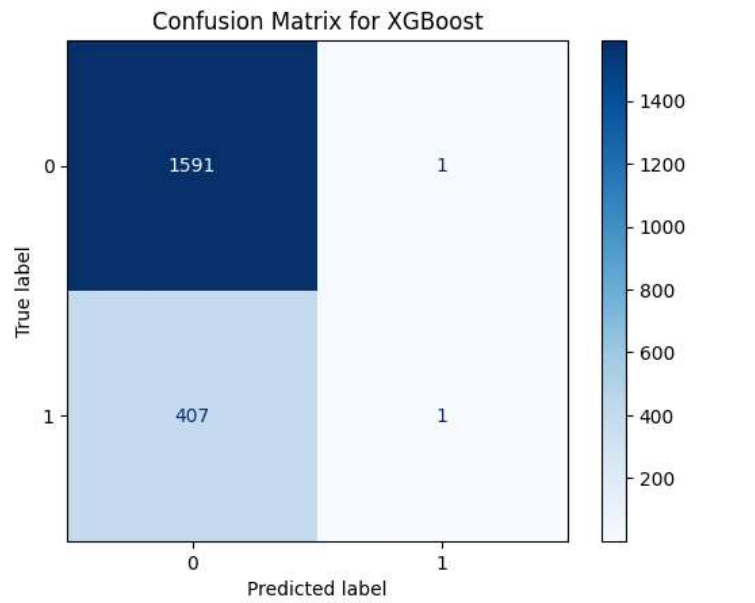


Fig. 6. Accuracy Comparison of Machine Learning Models

E. Feature Importance Analysis

Feature importance analysis was performed to identify the most influential factors affecting employee attrition. The analysis revealed that job satisfaction, promotion gap, monthly income, work stress, and age are the most significant features contributing to employee turnover prediction.

Employees with lower job satisfaction, higher work stress, and larger promotion gaps were found to have a higher probability of attrition. These insights can help organizations take preventive measures to improve employee retention.

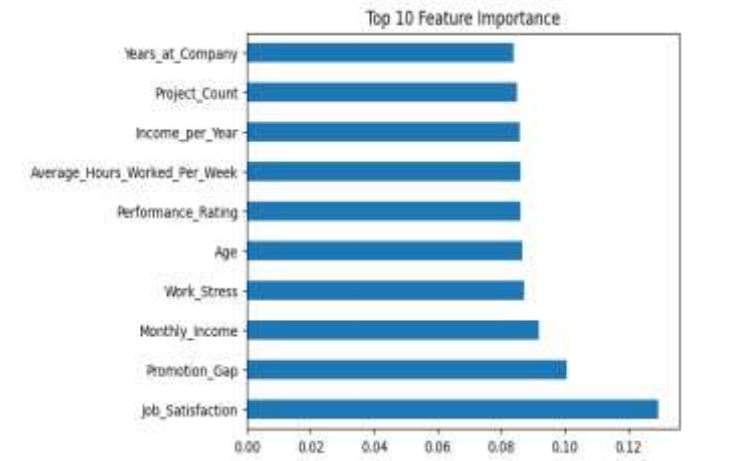


Fig. 7. Top 10 Feature Importance

F. Real-Time Prediction System

To improve practical usability, the trained XGBoost model

was deployed using a Streamlit-based web application. The application allows users to enter employee-related information and receive instant attrition predictions.

The real-time prediction system demonstrates the practical applicability of the proposed methodology in organizational HR environments. It can assist HR departments in identifying employees at risk of leaving the organization and support data-driven decision-making processes.

VI. CONCLUSION

This study successfully developed a machine learning-based employee attrition prediction system using a structured HR dataset containing 10,000 employee records. The proposed methodology incorporated data preprocessing, feature engineering, SMOTE-based class balancing, and multiple machine learning algorithms including Logistic Regression, Random Forest, and XGBoost. Among the implemented models, XGBoost achieved the highest prediction accuracy of 79.6%, demonstrating its effectiveness in handling complex employee behavior patterns. Feature importance analysis identified job satisfaction, promotion gap, monthly income, and work stress as major factors influencing employee attrition. In addition to predictive modeling, a Streamlit-based real-time prediction system was developed to improve practical usability in organizational environments. Although the achieved accuracy is slightly lower compared to some existing studies, the proposed system provides a balanced approach by combining prediction capability, interpretability, and real-time deployment. In future work, the system can be enhanced by integrating deep learning techniques, explainable AI methods such as SHAP, real-time organizational HR databases, and cloud-based deployment platforms to improve scalability, interpretability, and overall prediction performance.

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